Broder Card-Marking Strong Stationary Time
Proposition 8.6

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Private reading transcription of the fixed second-edition author PDF

Source and selected scope. *Markov Chains and Mixing Times, second edition*, by David A. Levin and Yuval Peres, "With contributions by Elizabeth L. Wilmer" as printed on physical page 1. The fixed author-hosted PDF has 461 physical pages and PDF generation metadata dated March 21, 2017; a separate author revision date is not stated on the inspected cover. The source is `https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf`. The sole primary is Proposition 8.6 and its complete printed proof on physical pages 121–122, printed pages 105–106. The stacking paragraph is retained as the author's heuristic. Definitions, the transition rule, Remark 8.7, complete Lemma 8.9 and Corollary 8.10, and the selected Notes are uncounted context or consequences. This transcription remains private.

Selected stopping-time definitions and mixing convention

Source: Section 6.2, physical page 93, printed page 77.

Let $\{\mathcal{F}_t\}$ be a *filtration*, a sequence of σ-algebras such that $\mathcal{F}_t \subset \mathcal{F}_{t+1}$ for all t. We say that $\{X_t\}$ is *adapted* to $\{\mathcal{F}_t\}$ if X_t is $\mathcal{F}_t$-measurable for all t. If $\mathcal{H}_t = \sigma(X_0, X_1, \ldots, X_t)$, then $\{\mathcal{H}_t\}$ is called the *natural filtration*. Clearly $\{X_t\}$ is adapted to the natural filtration.

Suppose that $\{X_t\}$ is adapted to $\{\mathcal{F}_t\}$. We say that $\{X_t\}$ is a *Markov chain with respect to* $\{\mathcal{F}_t\}$ if

$$\mathbf{P}_x\{X_{t+1} = y \mid \mathcal{F}_t\} = P(X_t, y), \tag{6.1}$$

where P is a transition matrix. A Markov chain as defined by (1.1) satisfies (6.1) when $\{\mathcal{F}_t\}$ is the natural filtration.

A *stopping time* for the filtration $\{\mathcal{F}_t\}$ is a random variable τ with values in $\{0, 1, \ldots\}$ such that $\{\tau = t\} \in \mathcal{F}_t$. If the filtration of a stopping time is not specified, it will be assumed to be the natural filtration.

Source: Section 6.4, physical page 94, printed page 78; strong stationary time definition only.

Let (X_t) be a Markov chain with respect to the filtration $\{\mathcal{F}_t\}$, with stationary distribution π. A *strong stationary time* for (X_t) and starting position x is an $\{\mathcal{F}_t\}$-stopping time τ, such that for all times t and all y,

$$\mathbf{P}_x\{\tau = t, X_\tau = y\} = \mathbf{P}_x\{\tau = t\}\pi(y). \tag{6.3}$$

In words, X_τ has distribution π and is independent of τ.

Source: Proposition 6.11, physical page 95, printed page 79; statement used as an uncounted input.

Proposition 6.11. *If τ is a strong stationary time for starting state x, then*

$$\|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} \le \mathbf{P}_x\{\tau > t\}. \tag{6.5}$$

Source: Selected conventions from Sections 4.4–4.5, physical pages 69–70, printed pages 53–54.

It is therefore convenient to define

$$d(t) := \max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|_{\mathrm{TV}}. \tag{4.22}$$

The mixing time is defined by

$$t_{\mathrm{mix}}(\varepsilon) := \min\{t : d(t) \le \varepsilon\} \tag{4.30}$$

and

$$t_{\mathrm{mix}} := t_{\mathrm{mix}}(1/4). \tag{4.31}$$

The random transposition shuffle

Source: Section 8.1, physical page 116, printed page 100; selected convention.

A probability distribution μ on the symmetric group describes a mechanism for shuffling cards: apply permutation σ to the deck with probability $\mu(\sigma)$. Repeatedly shuffling the deck using this mechanism is equivalent to running the random walk on the group with increment distribution μ. As discussed in Section 2.6, as long as the support of μ generates all of S_n, the resulting chain is irreducible. If $\mu(\mathrm{id}) > 0$, then it is aperiodic. Every shuffle chain has uniform stationary distribution.

Source: Section 8.2, physical page 118, printed page 102; complete transition rule and equation (8.3).

To avoid periodicity, the random transposition shuffle is defined as follows: At time t, choose two cards, labelled L_t and R_t, independently and uniformly at random. If L_t and R_t are different, transpose them. Otherwise, do nothing. The resulting distribution μ satisfies

$$\mu(\sigma) = \begin{cases} 1/n & \text{if } \sigma = \mathrm{id}, \\ 2/n^2 & \text{if } \sigma = (ij), \\ 0 & \text{otherwise.} \end{cases} \tag{8.3}$$

As in Example 8.2, this chain is irreducible; aperiodicity follows from Remark 8.3.

Editorial note. The earlier Section 8.2 lower-bound discussion says, "Consider the random transposition shuffle started from id." The primary proposition below is transcribed with its own printed initialization: no cards have been marked at time 0. No additional claim about an arbitrary random initial law is inserted.

8.2.3. Upper bound via strong stationary time

Source: Physical page 121, printed page 105; complete Proposition 8.6.

Proposition 8.6. *In the random transposition shuffle, let R_t and L_t be the cards chosen by the right and left hands, respectively, at time t. Assume that when $t = 0$, no cards have been marked. At time t, mark card R_t if both of the following are true:*

- R_t is unmarked.
- Either L_t is a marked card or $L_t = R_t$.

Let τ be the time when every card has been marked. Then τ is a strong stationary time for this chain.

Here is a heuristic explanation for why the scheme described above should give a strong stationary time. One way to generate a uniform random permutation is to build a stack of cards, one at a time, inserting each card into a uniformly random position relative to the cards already in the stack. For the stopping time described above, the marked cards are carrying out such a process.

Source: Physical pages 121–122, printed pages 105–106; complete printed proof, stopping before Remark 8.7.

Proof. It is clear that τ is a stopping time. To show that it is a strong stationary time, we prove the following subclaim by induction on t. Let $V_t \subseteq [n]$ be the set of cards marked at or before time t, and let $U_t \subseteq [n]$ be the set of positions occupied by V_t after the t-th transposition. We claim that given t, V_t, and U_t, all possible permutations of the cards in V_t on the positions U_t are equally likely.

This is clearly true when $t = 1$ (and continues to be true as long as at most one card has been marked).

Now, assume that the subclaim is true for t. The shuffler chooses cards L_{t+1} and R_{t+1}.

- If no new card is marked, then $V_{t+1} = V_t$. This can happen in three ways:
  - The cards L_{t+1} and R_{t+1} are different and both are unmarked. Then V_{t+1} and U_{t+1} are identical to V_t and U_t, respectively.
  - If L_{t+1} and R_{t+1} were both marked at an earlier round, then $U_{t+1} = U_t$ and the shuffler applies a uniform random transposition to the cards in V_t. All permutations of V_t remain equiprobable.
  - Otherwise, L_{t+1} is unmarked and R_{t+1} was marked at an earlier round. To obtain the position set U_{t+1}, we delete the position (at time t) of R_{t+1} and add the position (at time t) of L_{t+1}. For a fixed set U_t, all choices of $R_{t+1} \in U_t$ are equally likely, as are all permutations of V_t on U_t. Hence, once the positions added and deleted are specified, all permutations of V_t on U_{t+1} are equally likely.
- If the card R_{t+1} gets marked, then L_{t+1} is equally likely to be any element of $V_{t+1} = V_t \cup \{R_{t+1}\}$, while U_{t+1} consists of U_t along with the position of R_{t+1} (at time t). Specifying the permutation of V_t on U_t and the card L_{t+1} uniquely determines the permutation of V_{t+1} on U_{t+1}. Hence all such permutations are equally likely.

In every case, the collection of all permutations of the cards V_t on a specified set U_t together make equal contributions to all possible permutations of V_{t+1} on U_{t+1}. Hence, to conclude that all possible permutations of a fixed V_{t+1} on a fixed U_{t+1} are equally likely, we simply sum over all possible preceding configurations. ■

Source: Physical page 122, printed page 106; complete Remark 8.7.

Remark 8.7. In the preceding proof, the two subcases of the inductive step for which no new card is marked are essentially the same as checking that the uniform distribution is stationary for the random transposition shuffle and the random-to-top shuffle, respectively.

Editorial note. The primary proof preserves the printed phrase $R_{t+1} \in U_t$, while the preceding definition calls V_t the card set and U_t the position set. This notation is retained literally. The original base-case and predecessor-summation abbreviations have not been expanded.

Complete moment and mixing consequences, uncounted

Source: Physical pages 122–123, printed pages 106–107; complete Lemma 8.9 and its printed proof.

Lemma 8.9. *The stopping time τ defined in Proposition 8.6 satisfies*

$$\mathbf{E}(\tau) = 2n(\log n + O(1))$$

and

$$\mathrm{Var}(\tau) = O(n^2).$$

Proof. As for the coupon collector time, we can decompose

$$\tau = \tau_0 + \cdots + \tau_{n-1},$$

where τ_k is the number of transpositions after the k-th card is marked, up to and including when the $(k+1)$-st card is marked. The rules specified in Proposition 8.6 imply that τ_k is a geometric random variable with success probability $\frac{(k+1)(n-k)}{n^2}$ and that the τ_i's are independent of each other. Hence

$$\mathbf{E}(\tau) = \sum_{k=0}^{n-1} \frac{n^2}{(k+1)(n-k)}.$$

Substituting the partial fraction decomposition

$$\frac{1}{(k+1)(n-k)} = \frac{1}{n+1}\left[\frac{1}{k+1} + \frac{1}{n-k}\right]$$

and recalling that

$$\sum_{j=1}^{n} \frac{1}{j} = \log n + O(1)$$

(see Exercise 2.4) completes the estimate.

Now, for the variance. We can immediately write

$$\mathrm{Var}(\tau) = \sum_{k=0}^{n-1} \frac{1 - \frac{(k+1)(n-k)}{n^2}}{\left(\frac{(k+1)(n-k)}{n^2}\right)^2} < \sum_{k=0}^{n-1} \frac{n^4}{(k+1)^2(n-k)^2}.$$

Split the sum into two pieces:

$$\begin{aligned}\mathrm{Var}(\tau) &< \sum_{0 \le k < n/2} \frac{n^4}{(k+1)^2(n-k)^2} + \sum_{n/2 \le k < n} \frac{n^4}{(k+1)^2(n-k)^2} \\ &< \frac{2n^4}{(n/2)^2} \sum_{0 \le k \le n/2} \frac{1}{(k+1)^2} = O(n^2).\end{aligned}$$

■

Source: Physical page 123, printed page 107; complete Corollary 8.10 and its printed proof.

Corollary 8.10. *For the random transposition chain on an n-card deck,*

$$t_{\mathrm{mix}} \le (2 + o(1))n \log n.$$

Proof. Let τ be the Broder stopping time defined in Proposition 8.6, and let $t_0 = \mathbf{E}(\tau) + 2\sqrt{\mathrm{Var}(\tau)}$. By Chebyshev's inequality,

$$\mathbf{P}\{\tau > t_0\} \leq \frac{1}{4}.$$

Lemma 8.9 and Proposition 6.11 now imply the desired inequality. ■

Source: Selected original Notes, physical page 128, printed page 112.

The strong stationary time defined in Proposition 8.6 and used to prove the upper bound on the mixing time for random transpositions (Corollary 8.10) is due to A. Broder (see Diaconis (1988a)). This upper bound is off by a factor of 4. Matthews (1988b) gives an improved strong stationary time whose upper bound matches the lower bound.

Editorial note. These are the book's attribution and comparison statements. The book authors provide the collected exposition; Broder's route credit is kept separately. The later Matthews construction and its unexpanded analysis are outside this selected primary. No historical article or monograph has been newly acquired for this transcription.